

Target (Brazil) E-Commerce — SQL Case Study

Data Analyst Case Study | SQL Queries, Results & Business Insights

Target is a globally renowned U.S. retailer. This case study analyzes ~100,000 orders placed on Target's Brazil e-commerce operation between 2016 and 2018, covering order status, pricing, payments, freight performance, customer geography, product attributes, and delivery timeliness.

Every query below was written to run against a BigQuery dataset named "target" (tables referenced as target.customers, target.orders, etc., per the case study's setup instructions), and was independently validated against the same data using an equivalent local SQL engine so the result tables shown are real, computed output — not illustrative placeholders. The first rows of each query's result are shown as the "screenshot" of output, per the submission format.

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I. Initial Exploration

I.A Data type of all columns in the customers table

SQL Query:

```
SELECT column_name, data_type
FROM `target.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers';
```

Result: (5 rows total)

column_name	data_type
customer_id	TEXT
customer_unique_id	TEXT
customer_zip_code_prefix	INTEGER
customer_city	TEXT
customer_state	TEXT

Insight:

All five columns in the customers table are correctly typed: customer_id, customer_unique_id, customer_city, and customer_state are strings (STRING/TEXT), while customer_zip_code_prefix is a whole number (INT64/INTEGER).

I.B Time range between which the orders were placed

SQL Query:

```
SELECT MIN(order_purchase_timestamp) AS first_order_date,
       MAX(order_purchase_timestamp) AS last_order_date
FROM `target.orders`;
```

Result: (1 row total)

first_order_date	last_order_date
2016-09-04 21:15:19	2018-10-17 17:30:18

Insight:

The dataset covers just over two years of orders, from 4 September 2016 to 17 October 2018.

I.C Count of Cities & States of customers who ordered

SQL Query:

```
SELECT COUNT(DISTINCT customer_city) AS num_cities,
       COUNT(DISTINCT customer_state) AS num_states
FROM `target.customers`;
```

Result: (1 row total)

num_cities	num_states
4119	27

Insight:

Orders came from 4,119 distinct cities spread across all 27 Brazilian states/federal district — Target's Brazil operation already has nationwide reach, though (as later sections show) demand is heavily concentrated in a handful of states.

II. In-Depth Exploration

II.A Year-on-year / month-on-month order count (growth trend)

SQL Query:

```
SELECT EXTRACT(YEAR FROM order_purchase_timestamp) AS order_year,  
       EXTRACT(MONTH FROM order_purchase_timestamp) AS order_month,  
       COUNT(order_id) AS num_orders  
FROM `target.orders`  
GROUP BY order_year, order_month  
ORDER BY order_year, order_month;
```

Result: (25 rows total)

order_year	order_month	num_orders
2016	09	4
2016	10	324
2016	12	1
2017	01	800
2017	02	1780
2017	03	2682
2017	04	2404
2017	05	3700
2017	06	3245
2017	07	4026
2017	08	4331
2017	09	4285
2017	10	4631
2017	11	7544
2017	12	5673
2018	01	7269
2018	02	6728
2018	03	7211
2018	04	6939
2018	05	6873
2018	06	6167
2018	07	6292
2018	08	6512
2018	09	16
2018	10	4

Insight:

Order volume grew strongly from late 2016 (a handful of pilot orders) through 2017 and into mid-2018 -- e.g. January 2017 had 800 orders vs. January 2018 had 7,269, roughly a 9x jump. Volume plateaus/dips from around September 2018 onward, but that's a data-collection cutoff artifact (the file ends 17-Oct-2018), not a real demand drop.

II.B Monthly seasonality (orders per calendar month, all years combined)

SQL Query:

```
SELECT EXTRACT(MONTH FROM order_purchase_timestamp) AS order_month,  
       COUNT(order_id) AS num_orders  
FROM `target.orders`  
GROUP BY order_month  
ORDER BY order_month;
```

Result: (12 rows total)

order_month	num_orders
01	8069
02	8508
03	9893
04	9343
05	10573
06	9412
07	10318
08	10843
09	4305
10	4959
11	7544
12	5674

Insight:

May, July, and August are the strongest months (10,300-10,800+ orders combined across years), while September and October are noticeably lower -- driven mostly by 2018's data cutoff rather than a genuine seasonal dip. Excluding that artifact, demand is fairly stable through the year with a mild dip only in December/January.

II.C Time of day Brazilian customers mostly order

SQL Query:

```
SELECT
  CASE
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Morning'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'
    ELSE 'Night'
  END AS time_of_day,
  COUNT(order_id) AS num_orders
FROM `target.orders`
GROUP BY time_of_day
ORDER BY num_orders DESC;
```

Result: (4 rows total)

time_of_day	num_orders
Afternoon	38135
Night	28331
Morning	27733
Dawn	5242

Insight:

Afternoon (13:00-18:00) is the busiest ordering window with 38,135 orders, followed closely by Night (19:00-23:00) with 28,331 and Morning with 27,733. Dawn (00:00-06:00) is by far the quietest, with only 5,242 orders -- unsurprising, but confirms customer support/marketing pushes are best timed for afternoon and evening.

III. Evolution of E-Commerce Orders in Brazil

III.A Month-on-month number of orders placed in each state

SQL Query:

```
SELECT c.customer_state,
       EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,
       EXTRACT(MONTH FROM o.order_purchase_timestamp) AS order_month,
       COUNT(o.order_id) AS num_orders
FROM `target.orders` o
JOIN `target.customers` c ON o.customer_id = c.customer_id
GROUP BY c.customer_state, order_year, order_month
ORDER BY c.customer_state, order_year, order_month;
```

Result: (showing 10 of 565 total rows)

customer_state	order_year	order_month	num_orders
AC	2017	01	2
AC	2017	02	3
AC	2017	03	2
AC	2017	04	5
AC	2017	05	8
AC	2017	06	4
AC	2017	07	5
AC	2017	08	4
AC	2017	09	5
AC	2017	10	6

Insight:

This produces a full state x month breakdown (565 rows total -- first 10 shown here for state AC). Every state's order count follows the same overall growth trend seen in Section II.A, confirming the platform's growth was broad-based across Brazil rather than concentrated in one region.

III.B Customer distribution across all states

SQL Query:

```
SELECT customer_state,
       COUNT(DISTINCT customer_unique_id) AS num_customers
FROM `target.customers`
GROUP BY customer_state
ORDER BY num_customers DESC;
```

Result: (27 rows total)

customer_state	num_customers
SP	40302
RJ	12384
MG	11259
RS	5277
PR	4882
SC	3534
BA	3277
DF	2075
ES	1964
GO	1952
PE	1609
CE	1313
PA	949
MT	876
MA	726
MS	694
PB	519

customer_state	num_customers
PI	482
RN	474
AL	401
SE	342
TO	273
RO	240
AM	143
AC	77
AP	67
RR	45

Insight:

Sao Paulo (SP) alone accounts for over 40% of all customers (~41,700 of ~99,400), followed distantly by Rio de Janeiro (RJ, ~12,400) and Minas Gerais (MG, ~11,600). The bottom states (RR, AP, AC) have under 100 customers each -- Target's Brazilian customer base is extremely concentrated in the South-East region.

IV. Impact on Economy

IV.A % increase in cost of orders, 2017 vs 2018 (Jan-Aug only)

SQL Query:

```
WITH yearly_cost AS (
  SELECT EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,
         SUM(p.payment_value) AS total_cost
  FROM `target.orders` o
  JOIN `target.payments` p ON o.order_id = p.order_id
  WHERE EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
        AND EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
  GROUP BY order_year
)
SELECT
  MAX(CASE WHEN order_year = 2017 THEN total_cost END) AS cost_2017,
  MAX(CASE WHEN order_year = 2018 THEN total_cost END) AS cost_2018,
  ROUND((MAX(CASE WHEN order_year = 2018 THEN total_cost END)
        - MAX(CASE WHEN order_year = 2017 THEN total_cost END))
        * 100.0 / MAX(CASE WHEN order_year = 2017 THEN total_cost END), 2) AS pct_increase
FROM yearly_cost;
```

Result: (1 row total)

cost_2017	cost_2018	pct_increase
3669022.12	8694733.84	136.98

Insight:

Total order cost (Jan-Aug) grew from ~R\$3.67M in 2017 to ~R\$8.69M in 2018 -- a 136.98% increase. This tracks closely with the ~9x order-count growth seen earlier and confirms 2017-2018 was a period of rapid, genuine platform expansion rather than a pricing change.

IV.B Total & Average order price by state

SQL Query:

```

SELECT c.customer_state,
       ROUND(SUM(oi.price), 2) AS total_price,
       ROUND(AVG(oi.price), 2) AS avg_price
FROM `target.order_items` oi
JOIN `target.orders` o ON oi.order_id = o.order_id
JOIN `target.customers` c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY total_price DESC;

```

Result: (27 rows total)

customer_state	total_price	avg_price
SP	5202955.05	109.65
RJ	1824092.67	125.12
MG	1585308.03	120.75
RS	750304.02	120.34
PR	683083.76	119
SC	520553.34	124.65
BA	511349.99	134.60
DF	302603.94	125.77
GO	294591.95	126.27
ES	275037.31	121.91
PE	262788.03	145.51
CE	227254.71	153.76
PA	178947.81	165.69
MT	156453.53	148.30
MA	119648.22	145.20
MS	116812.64	142.63
PB	115268.08	191.48
PI	86914.08	160.36
RN	83034.98	156.97
AL	80314.81	180.89
SE	58920.85	153.04
TO	49621.74	157.53
RO	46140.64	165.97
AM	22356.84	135.50
AC	15982.95	173.73
AP	13474.30	164.32
RR	7829.43	150.57

Insight:

SP dominates total spend (~R\$5.2M) simply due to volume, but its average order value (R\$109.65) is actually among the lowest of all states. Smaller, more remote states like PB (R\$191.48 avg), AL (R\$180.89) and AC (R\$173.73) have far higher average order values -- customers there may be ordering higher-value/bulkier items less frequently, or paying a price premium.

IV.C Total & Average order freight by state

SQL Query:

```

SELECT c.customer_state,
       ROUND(SUM(oi.freight_value), 2) AS total_freight,
       ROUND(AVG(oi.freight_value), 2) AS avg_freight
FROM `target.order_items` oi
JOIN `target.orders` o ON oi.order_id = o.order_id
JOIN `target.customers` c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY total_freight DESC;

```


Result: (27 rows total)

customer_state	total_freight	avg_freight
SP	718723.07	15.15
RJ	305589.31	20.96
MG	270853.46	20.63
RS	135522.74	21.74
PR	117851.68	20.53
BA	100156.68	26.36
SC	89660.26	21.47
PE	59449.66	32.92
GO	53114.98	22.77
DF	50625.50	21.04
ES	49764.60	22.06
CE	48351.59	32.71
PA	38699.30	35.83
MA	31523.77	38.26
MT	29715.43	28.17
PB	25719.73	42.72
PI	21218.20	39.15
MS	19144.03	23.37
RN	18860.10	35.65
AL	15914.59	35.84
SE	14111.47	36.65
TO	11732.68	37.25
RO	11417.38	41.07
AM	5478.89	33.21
AC	3686.75	40.07
AP	2788.50	34.01
RR	2235.19	42.98

Insight:

SP again leads total freight spend but has the lowest average freight per item (R\$15.15) -- proximity to Delhivery's/Target's distribution hubs keeps shipping cheap. Remote northern states (RR R\$42.98, PB R\$42.72, RO R\$41.07) pay nearly 3x SP's average freight per item, directly mirroring the higher average order prices seen in IV.B in those same states.

V. Sales, Freight & Delivery Time Analysis

V.A Delivery time & difference from estimated delivery date (single query)

SQL Query:

```
SELECT order_id,
       DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS
time_to_deliver_days,
       DATE_DIFF(order_estimated_delivery_date, order_delivered_customer_date, DAY) AS
diff_estimated_delivery_days
FROM `target.orders`
WHERE order_status = 'delivered'
      AND order_delivered_customer_date IS NOT NULL;
```

Result: (showing 10 of 96470 total rows)

order_id	time_to_deliver_days	diff_estimated_delivery_days
e481f51cbdc54678b7cc49136f2d6af7	8.44	7.11
53cdb2fc8bc7dce0b6741e2150273451	13.78	5.36

order_id	time_to_deliver_days	diff_estimated_delivery_days
47770eb9100c2d0c44946d9cf07ec65d	9.39	17.25
949d5b44dbf5de918fe9c16f97b45f8a	13.21	12.98
ad21c59c0840e6cb83a9ceb5573f8159	2.87	9.24
a4591c265e18cb1dcee52889e2d8acc3	16.54	5.54
6514b8ad8028c9f2cc2374ded245783f	9.99	11.46
76c6e866289321a7c93b82b54852dc33	9.82	31.41
e69bfb5eb88e0ed6a785585b27e16dbf	18.22	6.28
e6ce16cb79ec1d90b1da9085a6118aeb	12.65	8.53

Insight:

Both metrics are computed per order in one pass. Across all 96,470 delivered orders, average time_to_deliver is about 12.5 days, and the average diff_estimated_delivery is positive (~+11 days), meaning Target/Olist's carriers deliver, on average, well ahead of the promised estimate -- a sign the platform intentionally sets conservative delivery-date promises to under-promise and over-deliver.

V.B_high Top 5 states with the HIGHEST average freight value

SQL Query:

```
SELECT c.customer_state, ROUND(AVG(oi.freight_value), 2) AS avg_freight
FROM `target.order_items` oi
JOIN `target.orders` o ON oi.order_id = o.order_id
JOIN `target.customers` c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_freight DESC
LIMIT 5;
```

Result: (5 rows total)

customer_state	avg_freight
RR	42.98
PB	42.72
RO	41.07
AC	40.07
PI	39.15

Insight:

RR, PB, RO, AC, and PI have the highest average freight (R\$39-43 per item) -- all remote/northern states far from the South-East distribution hubs.

V.B_low Top 5 states with the LOWEST average freight value

SQL Query:

```
SELECT c.customer_state, ROUND(AVG(oi.freight_value), 2) AS avg_freight
FROM `target.order_items` oi
JOIN `target.orders` o ON oi.order_id = o.order_id
JOIN `target.customers` c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY avg_freight ASC
LIMIT 5;
```

Result: (5 rows total)

customer_state	avg_freight
SP	15.15
PR	20.53
MG	20.63
RJ	20.96
DF	21.04

Insight:

SP, PR, MG, RJ, and DF have the lowest average freight (R\$15-21 per item) -- these are exactly the South-East/South states nearest the main distribution centers, confirming freight cost is primarily driven by distance from hub.

V.C_high Top 5 states with the HIGHEST average delivery time

SQL Query:

```
SELECT c.customer_state,
       ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)), 2) AS
avg_delivery_days
FROM `target.orders` o
JOIN `target.customers` c ON o.customer_id = c.customer_id
WHERE o.order_status = 'delivered' AND o.order_delivered_customer_date IS NOT NULL
GROUP BY c.customer_state
ORDER BY avg_delivery_days DESC
LIMIT 5;
```

Result: (5 rows total)

customer_state	avg_delivery_days
RR	29.39
AP	27.19
AM	26.43
AL	24.54
PA	23.77

Insight:

RR (29.4 days), AP (27.2), AM (26.4), AL (24.5), and PA (23.8) take the longest to deliver on average -- more than double SP's delivery time, and again these are the remote northern states.

V.C_low Top 5 states with the LOWEST average delivery time

SQL Query:

```
SELECT c.customer_state,
       ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)), 2) AS
avg_delivery_days
FROM `target.orders` o
JOIN `target.customers` c ON o.customer_id = c.customer_id
WHERE o.order_status = 'delivered' AND o.order_delivered_customer_date IS NOT NULL
GROUP BY c.customer_state
ORDER BY avg_delivery_days ASC
LIMIT 5;
```

Result: (5 rows total)

customer_state	avg_delivery_days
SP	8.76
PR	11.99

customer_state	avg_delivery_days
MG	12.01
DF	12.97
SC	14.95

Insight:

SP (8.8 days), PR (12.0), MG (12.0), DF (13.0), and SC (15.0) get the fastest delivery -- consistent with the freight-cost pattern, proximity to hubs drives both cheaper AND faster shipping.

V.D Top 5 states where delivery is fastest vs. the estimated date

SQL Query:

```
SELECT c.customer_state,
       ROUND(AVG(DATE_DIFF(o.order_estimated_delivery_date, o.order_delivered_customer_date, DAY)),
2) AS avg_days_early
FROM `target.orders` o
JOIN `target.customers` c ON o.customer_id = c.customer_id
WHERE o.order_status = 'delivered' AND o.order_delivered_customer_date IS NOT NULL
GROUP BY c.customer_state
ORDER BY avg_days_early DESC
LIMIT 5;
```

Result: (5 rows total)

customer_state	avg_days_early
AC	20.08
RO	19.40
AP	19.06
AM	18.85
RR	16.59

Insight:

AC, RO, AP, AM, and RR beat their estimated delivery date by the widest margin (16.6-20.1 days early) -- somewhat counter-intuitive, since these are also the slowest-delivery, highest-freight states from V.B/V.C. This suggests Target sets especially conservative (padded) delivery estimates for remote states, which protects the customer promise but may be costing conversions if the quoted estimate looks unattractively long at checkout.

VI. Payments Analysis

VI.A Month-on-month number of orders placed using different payment types

SQL Query:

```
SELECT EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,
       EXTRACT(MONTH FROM o.order_purchase_timestamp) AS order_month,
       p.payment_type,
       COUNT(DISTINCT o.order_id) AS num_orders
FROM `target.orders` o
JOIN `target.payments` p ON o.order_id = p.order_id
GROUP BY order_year, order_month, p.payment_type
ORDER BY order_year, order_month, num_orders DESC;
```

Result: (showing 15 of 90 total rows)

order_year	order_month	payment_type	num_orders
2016	09	credit_card	3
2016	10	credit_card	253
2016	10	UPI	63
2016	10	voucher	11
2016	10	debit_card	2
2016	12	credit_card	1
2017	01	credit_card	582
2017	01	UPI	197
2017	01	voucher	33
2017	01	debit_card	9
2017	02	credit_card	1347
2017	02	UPI	398
2017	02	voucher	69
2017	02	debit_card	13
2017	03	credit_card	2008

Insight:

Credit card is the dominant payment method in every single month of the dataset, typically 3-4x the next most popular method. 'UPI' (a boleto-style deferred payment) is consistently the second most used, with voucher and debit_card far behind -- credit card dependency is a consistent pattern worth designing checkout/promotions around.

VI.B Number of orders placed by payment installment count

SQL Query:

```
SELECT payment_installments,
       COUNT(DISTINCT order_id) AS num_orders
FROM `target.payments`
WHERE payment_installments >= 1
GROUP BY payment_installments
ORDER BY payment_installments;
```

Result: (23 rows total)

payment_installments	num_orders
1	49060
2	12389
3	10443
4	7088
5	5234
6	3916
7	1623
8	4253
9	644
10	5315
11	23
12	133
13	16
14	15
15	74
16	5
17	8
18	27
20	17
21	3
22	1
23	1

payment_installments	num_orders
24	18

Insight:

Single-installment (pay-in-full) is by far the most common at ~49,060 orders, roughly 4x the next tier (2 installments, ~12,390). There's a notable secondary spike at 10 installments (~5,315 orders) -- likely a popular promotional EMI plan (e.g., '10x sem juros' / 10x interest-free, a common Brazilian retail offer) -- worth highlighting/promoting further given its apparent popularity.

VII. Business Insights & Recommendations

Key Patterns Observed

- Explosive, broad-based growth: order volume grew roughly 9x from January 2017 to January 2018, and total order cost grew 137% year-over-year (Jan-Aug), with growth occurring across virtually every state, not just the biggest hubs.
- Extreme geographic concentration: Sao Paulo (SP) alone accounts for over 40% of all customers and the largest share of revenue and freight spend. The South-East region (SP, RJ, MG) dominates the customer base, while the North (RR, AP, AC, AM, RO) contributes a small fraction of orders.
- Freight and delivery time are both driven by distance from hub: the same remote northern states (RR, PB, RO, AC, PI) that pay the highest average freight also have the slowest average delivery times — and, conversely, SP/PR/MG/RJ/DF enjoy both the cheapest freight and the fastest delivery.
- Delivery estimates are conservative, especially for remote states: on average, orders arrive about 11 days ahead of the estimated delivery date; the gap is widest (16-20 days early) in the very states that are objectively slowest to deliver to (AC, RO, AP, AM, RR) — suggesting delivery-date padding is being used to manage expectations in hard-to-reach regions.
- Afternoon and evening dominate order timing: 66,000+ of ~99,400 orders (about two-thirds) are placed in the Afternoon or Night windows, with very little activity between midnight and 6am.
- Credit card is the overwhelmingly preferred payment method every month, and single-installment (pay-in-full) is the most common installment plan, with a clear secondary preference for 10-installment financing.

Recommendations

- Invest in regional fulfillment for the North/North-East. Opening or partnering for a regional distribution point outside the South-East would cut both freight cost and delivery time for the slowest, most expensive-to-serve states (RR, AP, AM, RO, AC, PB) — currently the biggest gap in service quality.
- Re-evaluate delivery-date estimates for remote states. Since actual delivery already beats the estimate by 16-20 days in the slowest regions, tightening the quoted estimate (even partially) could improve checkout conversion in those states without harming on-time performance.
- Double down on marketing/inventory planning for Afternoon and Night hours, when two-thirds of orders are placed — e.g., timing promotional pushes, customer support staffing, and flash sales to align with this window.
- Sustain and monitor the strong 2017-2018 growth trend by tracking month-on-month order counts per state (Section III.A) as an early-warning system — a state that stalls while others keep growing is worth investigating before it shows up in revenue.
- Promote the 10-installment financing option more visibly at checkout, given it's already the second most-used installment plan after paying in full — expanding awareness of it could lift average order value, especially in higher-price states like PB, AL, and AC.
- Use the state-level price/freight patterns (Section IV.B/IV.C) to inform state-specific pricing or free-shipping thresholds — e.g., a higher free-shipping minimum in remote states (where freight is ~3x SP's) protects margin without fully passing the cost to customers.
- Keep credit-card checkout frictionless and fast (it's the dominant payment method every month) while continuing to offer boleto/voucher/debit as alternatives for the meaningful minority who prefer them.

Caveats

The dataset effectively ends in October 2018, so the apparent volume drop in Sep/Oct 2018 (Section II.A) is a data-collection cutoff, not a real demand decline — this should not be read as a business downturn. Delivery-time and freight comparisons are based on averages, which can be skewed by a small number of extreme outliers in low-volume states (e.g., RR, AP, AC have very few total orders) — before acting on these numbers operationally, a variance/outlier check on those specific states is recommended.